```
clc;
clear;
close all;

% Question 1
R1 = 10e3;
C1 = 100e-6;
tau1 = R1*C1;
G1 = tf(1, [tau1 1]);

% Question 2
R2 = 4.7e3;
C2 = 220e-6;
tau2 = R2*C2;
G2 = tf(1, [tau2 1]);

% Question 3
K3 = 2.5;
tau3 = 8/4;
G3 = tf(K3, [tau3 1]);

% Display transfer functions and results
disp('Question 1:');
G1
tau1
dcgain(G1)
stepinfo(G1)

disp('Question 2:');
G2
tau2
dcgain(G2)
stepinfo(G2)

disp('Question 3:');
G3
pole(G3)
dcgain(G3)
stepinfo(G3)

% Plot all three step responses
figure;
step(G1, G2, G3);
grid on;
legend('Question 1', 'Question 2', 'Question 3');
title('RC and First-Order System Step Responses');
xlabel('Time (s)');
ylabel('Output');
```

```
Question 1:

G1 =
```

```
    1
  -----
  s + 1

Continuous-time transfer function.
tau1 =
     1
ans =
     1
ans =
  struct with fields:

         RiseTime: 2.1970
    TransientTime: 3.9121
     SettlingTime: 3.9121
      SettlingMin: 0.9045
      SettlingMax: 0.9993
        Overshoot: 0
       Undershoot: 0
             Peak: 0.9993
         PeakTime: 7.3222
Question 2:

G2 =

       1
  -----------
  1.034 s + 1

Continuous-time transfer function.
tau2 =
    1.0340
ans =
     1
ans =
  struct with fields:

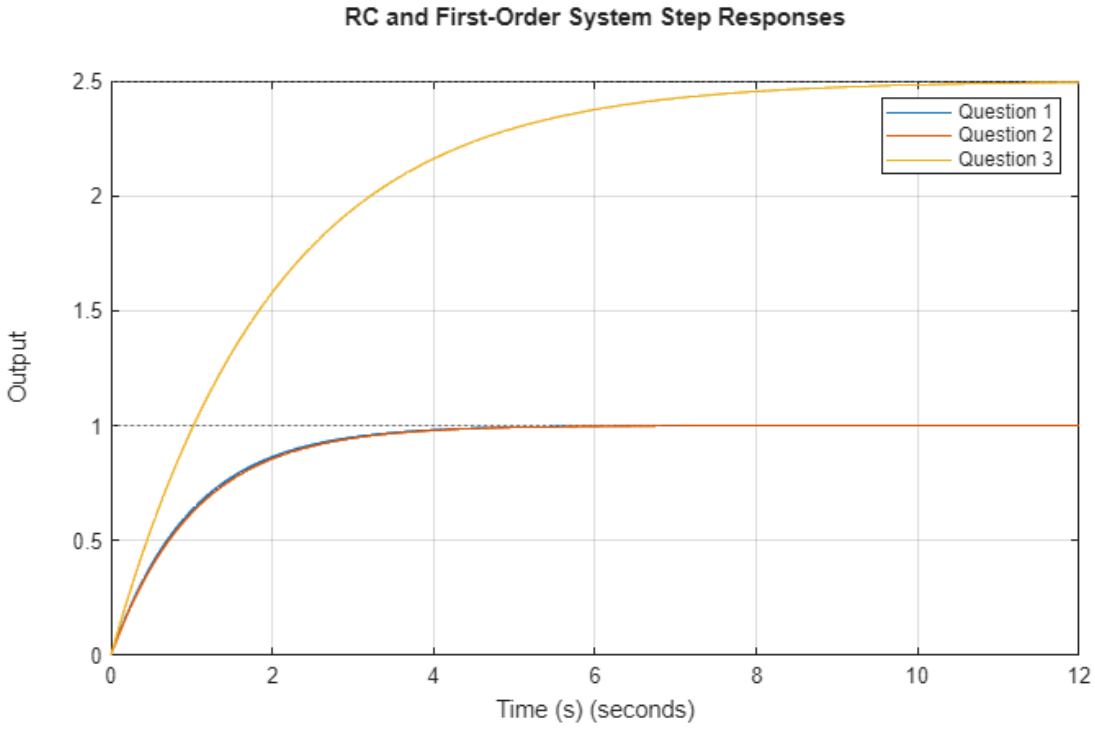
         RiseTime: 2.2717
    TransientTime: 4.0451
     SettlingTime: 4.0451
      SettlingMin: 0.9045
      SettlingMax: 0.9993
        Overshoot: 0
       Undershoot: 0
             Peak: 0.9993
         PeakTime: 7.5712
Question 3:

G3 =

    2.5
  -------
  2 s + 1
```

```
Continuous-time transfer function.
ans =
   -0.5000
ans =
    2.5000
ans =
  struct with fields:

         RiseTime: 4.3940
    TransientTime: 7.8241
     SettlingTime: 7.8241
      SettlingMin: 2.2613
      SettlingMax: 2.4983
        Overshoot: 0
       Undershoot: 0
             Peak: 2.4983
         PeakTime: 14.6444
```



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